THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The square of an even number is always: 1

(a) prime **(b)** a multiple of 3 **(c)** odd **(d)** even

2. The sum of the first 12 odd natural numbers is: 1

(a) 144 **(b)** 143 **(c)** 156 **(d)** 24

3. The number 6300000 in standard (scientific) form is: 1

(a) 6.3×10^6 **(b)** 6.3×10^7 **(c)** 63×10^5 **(d)** 6.3×10^5

4. Which Roman symbol stands for 500? 1

(a) L **(b)** C **(c)** M **(d)** D

5. Which of the following cannot have a pair of parallel sides? 1

(a) trapezium **(b)** kite (in general)
(c) rhombus **(d)** rectangle

6. The diagonals of a rhombus are 10 cm and 24 cm. Its side is: 1

(a) 14 cm **(b)** 12 cm **(c)** 13 cm **(d)** 17 cm

7. Is 8591 divisible by 11? 1

(a) Yes **(b)** No
(c) Only if it is odd **(d)** Only if it ends in 1

8. $(3 \times 4) \times 5 = 3 \times (4 \times 5)$ illustrates the: 1

(a) inverse property **(b)** associative property
(c) commutative property **(d)** distributive property

9. The ratio 28 : 35 in its simplest form is: 1

(a) 4 : 6 **(b)** 5 : 5 **(c)** 4 : 5 **(d)** 5 : 4

10. 750 g in kg is: 1

(a) 0.75 kg **(b)** 7.5 kg **(c)** 0.075 kg **(d)** 75 kg

11. If 20% of a number is 45, the number is: 1

(a) 180 **(b)** 225 **(c)** 270 **(d)** 90

12. An article bought for ₹ 1200 is sold for ₹ 1500. The profit percent is: 1

(a) 30% **(b)** 12.5% **(c)** 300% **(d)** 25%

13. The triplet obtained from 2, 1 using $(m^2 - n^2,\ 2mn,\ m^2 + n^2)$ is: 1

(a) (3, 4, 5) **(b)** (2, 3, 5) **(c)** (1, 2, 3) **(d)** (4, 5, 6)

14. Is a triangle with sides 9, 12 and 16 a right triangle? 1

(a) No **(b)** Cannot say
(c) Only if it is isosceles **(d)** Yes

15. On a map, 1 cm represents 12.5 km. A distance of 8 cm on the map represents: 1

(a) 200 km **(b)** 100 km **(c)** 112.5 km **(d)** 50 km

16. The number of edges of a triangular pyramid (tetrahedron) is: 1

(a) 6 **(b)** 9 **(c)** 5 **(d)** 7

17. The median of 16, 18, 20, 22, 24, 54 is: 1

(a) 20 **(b)** 28.6667 **(c)** 21 **(d)** 22

18. The result of the trick 'Think of n, double it, add 10, halve, subtract n' is: 1

(a) 0 **(b)** n **(c)** 5 **(d)** 10

19. The area of a square whose diagonal is 10 cm is: 1

(a) 25 cm² **(b)** 20 cm² **(c)** 100 cm² **(d)** 50 cm²

20. The area of a rectangle of length 24 cm and breadth 5 cm is: 1

(a) 120 cm² **(b)** 29 cm² **(c)** 125 cm² **(d)** 58 cm²

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Compare 3×10^4 and 2×10^5. 2

22. What is the role of zero in a place value system? 2

23. Find the digit x if $46x$ is divisible by 9. 2

24. Write the ratio of 2 hours to 40 minutes. 2

25. The sides of a rectangle are 20 m and 21 m. Find its diagonal. 2

26. A pie chart shows 300 students; 25% prefer cricket. How many? 2

27. If the mean of $x, x + 2, x + 4$ is 10, find x. 2

28. Find the area of a rhombus with each side 13 cm and one diagonal 10 cm. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. A cubical tank holds 13824 litres (1 m³ = 1000 L). Find the length of its edge. 3

30. In kite ABCD, AB = AD and CB = CD. If $\angle B = \angle D = 90^\circ$ and $\angle A = 70^\circ$, find $\angle C$. 3

31. A rectangle has length $(x + 5)$ and breadth $(x - 2)$. Find its area as an expression and at $x = 6$. 3

32. At what total amount will ₹ 9000 grow in 5 years at 4% simple interest? 3

33. Why can a regular pyramid with base 7 sides have 8 faces? 3

34. Show that 78 and its reverse add to a multiple of 11 and explain algebraically. 3

35. A rectangular park measures 9 m by 12 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

OR

Show that (2, 3, 4) is not a Pythagorean triplet while (9, 12, 15) is. 3

36. A rectangular carpet 6 m × 4 m is laid in a room 8 m × 6 m. Find the uncovered area. 3

OR

A triangle and a parallelogram are on the same base 10 cm and between the same parallels 6 cm apart. Compare their areas. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Why do scientists prefer scientific notation? 5
(ii) Write the mass of an electron, 0.00000000000000000000000000000091 kg, in scientific notation.

OR

(i) Prove $a^m \times a^{-m} = 1$. 5
(ii) Verify for $a = 3,\ m = 2$.

38. (i) Express the ratio 3 : 5 as percentages. 5
(ii) A mixture is 35% acid; how much acid in 80 L?

OR

(i) Find the principal that amounts to ₹ 5832 in 2 years at 8% CI. 5
(ii) State what is compounding.

39. (i) Find x if the median of 3, 5, x, 9, 11 is 8. 5
(ii) Find the mean.

OR

(i) Show that the sum of deviations of values from their mean is zero for 4, 6, 8, 10, 12. 5
(ii) Explain the balancing idea.

40. (i) Prove that parallelograms on the same base between the same parallels have equal area. 5
(ii) Illustrate with an example.

OR

(i) Find the area of a parallelogram with base 24 cm and height 15 cm. 5
(ii) If its slant side is 17 cm, find the area of the triangle cut off by the height.

END OF QUESTION PAPER — Total: 80 Marks

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